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About

Prof. Xu received his BEng degree in computer science and engineering from Zhejiang University, Hangzhou, China and his PhD degree in computer science from Hong Kong University of Science and Technology. With an H-index of 63, Prof. Xu has published over 300 refereed articles in top-tier international conferences and journals, including SIGMOD, VLDB, ICDE, TKDE, and VLDBJ. He has been invited to serve as Program Committee Co-Chair and Area Chair for several reputed international conferences, such as ICDCS '12, WAIM '16, APWeb-WAIM '18, MDM '19, NDBC '20, and ICDE '24. He was a recipient of HKBU President's Award for Outstanding Performance in Scholarly Work (2017) and HKBU President's Award for Outstanding Performance in Teaching (2018). He has served as an Associate Editor of IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Big Data (TBD), IEEE Transactions on Parallel and Distributed Systems (TPDS), Proceedings of VLDB Endowment (PVLDB), and Springer Journal of Data Science and Engineering (DSE). He is a Fellow of IEEE.

Research Interests

- Applied AI & LLM
- Databases
- Blockchain
- Data Security and Privacy

Selected Publications

- Y. Cao, Z. Gao, Z. Li, X. Xie, S. K. Zhou, and J. Xu. "LEGO-GraphRAG: Modularizing Graph-based Retrieval-Augmented Generation for Design Space Exploration." *Proc. the 51st International Conference on Very Large Data Bases (**PVLDB '25**)*, London, United Kingdom, 2025.
- X. Zhang, Q. Wang, C. Xu, Y. Peng, and J. Xu. "FedKNN: Secure Federated k-Nearest Neighbor Search." *Proc. the ACM SIGMOD Int'l Conf. on Management of Data (**SIGMOD '24**)*, Santiago, Chile, 2024.
- Y. Peng, B. Choi, T. N. Chan, J. Yang, and J. Xu. "Efficient Approximate Nearest Neighbor Search in Multi-dimensional Databases." *Proc. the ACM SIGMOD Int'l Conf. on Management of Data (**SIGMOD '23**)*, Seattle, WA, USA, 2023.
- T. N. Chan, P. L. Ip, L. H. U, B. Choi, and J. Xu. "SAFE: A Share-and-Aggregate Bandwidth Exploration Framework for Kernel Density Visualization." *Proc. the 47th International Conference on Very Large Data Bases (**PVLDB '22**)*, Sydney, Australia, 2022.
- X. Liao*, Q. Liu*, J. Jiang, X. Huang, J. Xu, and B. Choi. "Distributed D-core Decomposition over Large Directed Graphs." *Proc. the 47th International Conference on Very Large Data Bases (**PVLDB '22**)*, Sydney, Australia, 2022. (*equal contribution)

- T. N. Chan, L. H. U, B. Choi, and J. Xu. "SLAM: Efficient Sweep Line Algorithms for Kernel Density Visualization." *Proc. the ACM SIGMOD International Conference on Management of Data (**SIGMOD '22**)*, Philadelphia, USA, June 2022.
- C. Xu*, C. Zhang*, J. Xu, and J. Pei. "SlimChain: Scaling Blockchain Transactions through Off-Chain Storage and Parallel Processing." *Proc. the 47th International Conference on Very Large Data Bases (**PVLDB '21**)*, Copenhagen, Denmark, 2021. (*equal contribution)
- Q. Liu, X. Zhu, X. Huang, and J. Xu. "Local Algorithms for Distance-generalized Core Decomposition over Large Dynamic Graphs." *Proc. the 47th International Conference on Very Large Data Bases (**PVLDB '21**)*, Copenhagen, Denmark, 2021.
- Q. Liu, M. Zhao, X. Huang, J. Xu, and Y. Gao. "Truss-based Community Search over Large Directed Graphs." *Proc. the ACM SIGMOD International Conference on Management of Data (**SIGMOD '20**)*, Portland, OR, USA, June 2020.
- C. Xu, C. Zhang, and J. Xu. "vChain: Enabling Verifiable Boolean Range Queries over Blockchain Databases." *Proc. the ACM SIGMOD International Conference on Management of Data (**SIGMOD '19**)*, Amsterdam, Netherlands, 2019.